InternNova Internship

Week 2

Assignment Tasks

Name: Ayush Kumar

Batch Cohort: Batch-03

Student ID: INST26DA030238

Internship Domain: Data Analytics

Submission Date: 15/08/2026

Index

S. No.	Task	Topic
1	Task 1	NumPy Basics: Arrays, Shape, Size & Data Type
2	Task 2	NumPy Indexing, Slicing & Reshaping
3	Task 3	NumPy Mathematical & Statistical Operations
4	Task 4	Pandas Series & DataFrame
5	Task 5	Reading & Inspecting Data
6	Task 6	Selecting, Filtering & Sorting Data
7	Task 7	Handling Missing Values
8	Task 8	Merge, Concatenate, GroupBy & Pivot Table
9	Task 9	Exporting Data
10	Task 10	Mini Data Analysis Project

The GitHub repository

https://github.com/ayushkumar1311/InternNova-Internship-tasks-week-2

Q1.

```python
import numpy as np

arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90, 100])

print("Array:", arr)

print("Shape:", arr.shape)
print("Size:", arr.size)
print("Data Type:", arr.dtype)

one_d = np.array([1, 2, 3, 4, 5])
print("\nOne-Dimensional Array:")
print(one_d)

two_d = np.array([[1, 2, 3], [4, 5, 6]])
print("\nTwo-Dimensional Array:")
print(two_d)
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/NumPy Introduction & Arrays.py
Array: [ 10  20  30  40  50  60  70  80  90 100]
Shape: (10,)
Size: 10
Data Type: int32

One-Dimensional Array:
[1 2 3 4 5]

Two-Dimensional Array:
[[1 2 3]
 [4 5 6]]
>>>
```

Brief Explanation:

Creation and inspection of NumPy arrays, including their shape, size, and data type. One-dimensional and two-dimensional arrays are also created.

Q2.

```python
import numpy as np
arr = np.array([10, 20, 30, 40, 50, 60, 70, 80])

print("Element at index 2:", arr[2])
print("Element at index 5:", arr[5])

print("Sliced array:", arr[2:6])

two_d = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
])

print("\nTwo-Dimensional Array:")
print(two_d)

print("\nFirst Row:", two_d[0])
print("Second Row:", two_d[1])

print("First Column:", two_d[:, 0])
print("Second Column:", two_d[:, 1])

print("\nOriginal Array:")
print(arr)

reshaped = arr.reshape(2, 4)

print("\nReshaped Array (2 x 4):")
print(reshaped)

reshaped_2 = arr.reshape(4, 2)

print("\nReshaped Array (4 x 2):")
print(reshaped_2)
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/NumPy Indexing, Slicing & Reshaping.py
Element at index 2: 30
Element at index 5: 60
Sliced array: [30 40 50 60]

Two-Dimensional Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]

First Row: [1 2 3]
Second Row: [4 5 6]
First Column: [1 4 7]
Second Column: [2 5 8]

Original Array:
[10 20 30 40 50 60 70 80]

Reshaped Array (2 x 4):
[[10 20 30 40]
 [50 60 70 80]]

Reshaped Array (4 x 2):
[[10 20]
 [30 40]
 [50 60]
 [70 80]]
>>>
```

Brief Explanation:

Demonstrates accessing elements using indexing, extracting portions using slicing, accessing rows and columns, and reshaping arrays into different dimensions.

Q3.

```python
import numpy as np

a = np.array([10, 20, 30, 40, 50])
b = np.array([2, 4, 5, 8, 10])

print("Addition:", np.add(a, b))
print("Subtraction:", np.subtract(a, b))
print("Multiplication:", np.multiply(a, b))
print("Division:", np.divide(a, b))

print("Mean:", np.mean(a))
print("Median:", np.median(a))
print("Minimum:", np.min(a))
print("Maximum:", np.max(a))
print("Standard Deviation:", np.std(a))
print("Sum:", np.sum(a))
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/NumPy Mathematical & Statistical Operations.py
Addition: [12 24 35 48 60]
Subtraction: [ 8 16 25 32 40]
Multiplication: [ 20  80 150 320 500]
Division: [5. 5. 6. 5. 5.]
Mean: 30.0
Median: 30.0
Minimum: 10
Maximum: 50
Standard Deviation: 14.142135623730951
Sum: 150
>>> KeyboardInterrupt
>>>
```

Brief Explanation:

Performs addition, subtraction, multiplication, division, mean, median, minimum, maximum, standard deviation, and sum using NumPy.

Q4.

```python
import pandas as pd

series = pd.Series([85, 90, 78, 92, 88])

data = {
    "Name": ["Ayush", "Rahul", "Aman", "Priya", "Riya"],
    "Age": [21, 20, 22, 21, 20],
    "Marks": [85, 90, 78, 92, 88]
}

df = pd.DataFrame(data)

print("Pandas Series:")
print(series)

print("\nDataFrame:")
print(df)

print("\nColumn Names:")
print(df.columns)

print("\nIndex:")
print(df.index)

df["Grade"] = ["A", "A+", "B", "A+", "A"]

print("\nUpdated DataFrame:")
print(df)
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/Pandas Series & DataFrame.py
Pandas Series:
0    85
1    90
2    78
3    92
4    88
dtype: int64

DataFrame:
    Name  Age  Marks
0  Ayush   21     85
1  Rahul   20     90
2   Aman   22     78
3  Priya   21     92
4   Riya   20     88

Column Names:
Index(['Name', 'Age', 'Marks'], dtype='object')

Index:
RangeIndex(start=0, stop=5, step=1)

Updated DataFrame:
    Name  Age  Marks Grade
0  Ayush   21     85     A
1  Rahul   20     90    A+
2   Aman   22     78     B
3  Priya   21     92    A+
4   Riya   20     88     A
>>>
```

Brief Explanation:

Creates a Pandas Series and a DataFrame containing student information. The program displays columns, indexes, and adds a new column.

Q5.

students.csv

Name	Age	Gender	Marks	Grade
Ayush	21	Male	85	A
Rahul	20	Male	90	A+
Aman	22	Male	78	B
Priya	21	Female	92	A+
Neha	20	Female	88	A
Rohan	22	Male	75	B
Anjali	21	Female	95	A+
Vivek	20	Male	82	A
Sneha	22	Female	89	A
Karan	21	Male	70	B

Task 5 Reading & Inspecting Data.py - C:\Users\ASUS\Desktop\internnova\week 2\task 5\Task 5 Reading & Inspecting Data.py (3.12.10)

File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("students.csv")

print("First 5 Rows:")
print(df.head())

print("\nLast 5 Rows:")
print(df.tail())

print("\nNumber of Rows and Columns:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()

print("\nStatistical Summary:")
print(df.describe())
```

Ln: 1 Col: 0

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\ASUS\Desktop\internnova\week 2\task 5\Task 5 Reading & Inspecting Data.py
First 5 Rows:
     Name  Age  Gender  Marks Grade
0   Ayush   21    Male     85     A
1   Rahul   20    Male     90    A+
2    Aman   22    Male     78     B
3   Priya   21  Female     92    A+
4    Neha   20  Female     88     A

Last 5 Rows:
     Name  Age  Gender  Marks Grade
5   Rohan   22    Male     75     B
6  Anjali   21  Female     95    A+
7   Vivek   20    Male     82     A
8   Sneha   22  Female     89     A
9   Karan   21    Male     70     B

Number of Rows and Columns:
(10, 5)

Column Names:
Index(['Name', 'Age', 'Gender', 'Marks', 'Grade'], dtype='object')

Data Types:
Name      object
Age        int64
Gender    object
Marks      int64
Grade     object
dtype: object

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
Ln: 59  Col: 0
```

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
Column Names:
Index(['Name', 'Age', 'Gender', 'Marks', 'Grade'], dtype='object')

Data Types:
Name      object
Age        int64
Gender    object
Marks      int64
Grade     object
dtype: object

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    10 non-null     object
 1   Age     10 non-null     int64
 2   Gender  10 non-null     object
 3   Marks   10 non-null     int64
 4   Grade   10 non-null     object
dtypes: int64(2), object(3)
memory usage: 532.0+ bytes

Statistical Summary:
             Age      Marks
count  10.000000  10.000000
mean   21.000000  84.400000
std     0.816497   8.016649
min    20.000000  70.000000
25%    20.250000  79.000000
50%    21.000000  86.500000
75%    21.750000  89.750000
max    22.000000  95.000000
>>>
Ln: 59  Col: 0
```

Brief Explanation:

Reads a CSV dataset using Pandas and examines its rows, columns, data types, structure, and statistical summary.

Q6.

students1.csv

Name	Age	Gender	Marks	Grade
Ayush	21	Male	85	A
Rahul	20	Male	90	A+
Aman	22	Male	78	B
Priya	21	Female	92	A+
Neha	20	Female	88	A
Rohan	22	Male	75	B
Anjali	21	Female	95	A+
Vivek	20	Male	82	A
Sneha	22	Female	89	A
Karan	21	Male	70	B

Task 6 Selecting, Filtering & Sorting Data.py - C:/Users/ASUS/Desktop/internnova/week 2/Task 6 Selecting, Filtering & Sorting Data.py (3.12.10)

File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv(r"C:\Users\ASUS\Desktop\internnova\week 2\task 6\students1.csv")

print("Original DataFrame:")
print(df)

print("\nSelected Columns:")
print(df[["Name", "Marks"]])

print("\nSelected Rows:")
print(df.iloc[0:3])

print("\nStudents with Marks greater than 80:")
print(df[df["Marks"] > 80])

print("\nStudents with Marks greater than 80 and Age less than 22:")
print(df[(df["Marks"] > 80) & (df["Age"] < 22)])

print("\nSorted by Marks in Ascending Order:")
print(df.sort_values("Marks"))

print("\nSorted by Marks in Descending Order:")
print(df.sort_values("Marks", ascending=False))
```

Ln: 3 Col: 80

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/Task 6 Selecting, Filtering & Sorting Data.py
Original DataFrame:
     Name  Age  Gender  Marks Grade
0   Ayush   21    Male     85     A
1   Rahul   20    Male     90    A+
2    Aman   22    Male     78     B
3   Priya   21  Female     92    A+
4    Neha   20  Female     88     A
5   Rohan   22    Male     75     B
6  Anjali   21  Female     95    A+
7   Vivek   20    Male     82     A
8   Sneha   22  Female     89     A
9   Karan   21    Male     70     B

Selected Columns:
     Name  Marks
0   Ayush     85
1   Rahul     90
2    Aman     78
3   Priya     92
4    Neha     88
5   Rohan     75
6  Anjali     95
7   Vivek     82
8   Sneha     89
9   Karan     70

Selected Rows:
    Name  Age Gender  Marks Grade
0  Ayush   21   Male     85     A
1  Rahul   20   Male     90    A+
2   Aman   22   Male     78     B

Students with Marks greater than 80:
     Name  Age  Gender  Marks Grade
Ln: 81  Col: 0
```

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help

Students with Marks greater than 80 and Age less than 22:
     Name  Age  Gender  Marks Grade
0   Ayush   21    Male     85     A
1   Rahul   20    Male     90    A+
3   Priya   21  Female     92    A+
4    Neha   20  Female     88     A
6  Anjali   21  Female     95    A+
7   Vivek   20    Male     82     A

Sorted by Marks in Ascending Order:
     Name  Age  Gender  Marks Grade
9   Karan   21    Male     70     B
5   Rohan   22    Male     75     B
2    Aman   22    Male     78     B
7   Vivek   20    Male     82     A
0   Ayush   21    Male     85     A
4    Neha   20  Female     88     A
8   Sneha   22  Female     89     A
1   Rahul   20    Male     90    A+
3   Priya   21  Female     92    A+
6  Anjali   21  Female     95    A+

Sorted by Marks in Descending Order:
     Name  Age  Gender  Marks Grade
6  Anjali   21  Female     95    A+
3   Priya   21  Female     92    A+
1   Rahul   20    Male     90    A+
8   Sneha   22  Female     89     A
4    Neha   20  Female     88     A
0   Ayush   21    Male     85     A
7   Vivek   20    Male     82     A
2    Aman   22    Male     78     B
5   Rohan   22    Male     75     B
9   Karan   21    Male     70     B
>>>
Ln: 81  Col: 0
```

Brief Explanation:

Demonstrates selecting columns and rows, filtering records using conditions, applying multiple conditions, and sorting data in ascending and descending order.

Q7.

students_missing.csv

Name	Age	Gender	Marks	Grade
Ayush	21	Male	85	A
Rahul	20	Male		A+
Aman		Male	78	B
Priya	21	Female	92	
Neha	20	Female		A
Rohan		Male	75	B
Anjali	21	Female	95	A+
Vivek	20	Male	82	
Sneha	22	Female	89	A
Karan	21	Male		B

Task 7 Handling Missing Values.py - C:/Users/ASUS/Desktop/internnova/week 2/task 7/Task 7 Handling Missing Values.py (3.12.10)

File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("students_missing.csv")

print("Dataset Before Handling Missing Values:")
print(df)

print("\nMissing Values:")
print(df.isnull())

print("\nCount of Missing Values in Each Column:")
print(df.isnull().sum())

df_dropped = df.dropna()

print("\nDataset After Removing Rows with Missing Values:")
print(df_dropped)

df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
df["Grade"] = df["Grade"].fillna("Not Available")

print("\nDataset After Filling Missing Values:")
print(df)
```

Ln: 25 Col: 0

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/task 7/Task 7 Handling Missing Values.py
Dataset Before Handling Missing Values:
     Name   Age  Gender  Marks Grade
0   Ayush  21.0    Male   85.0     A
1   Rahul  20.0    Male    NaN    A+
2    Aman   NaN    Male   78.0     B
3   Priya  21.0  Female   92.0   NaN
4    Neha  20.0  Female    NaN     A
5   Rohan   NaN    Male   75.0     B
6  Anjali  21.0  Female   95.0    A+
7   Vivek  20.0    Male   82.0   NaN
8   Sneha  22.0  Female   89.0     A
9   Karan  21.0    Male    NaN     B

Missing Values:
    Name    Age  Gender  Marks  Grade
0  False  False   False  False  False
1  False  False   False   True  False
2  False   True   False  False  False
3  False  False   False  False   True
4  False  False   False   True  False
5  False   True   False  False  False
6  False  False   False  False  False
7  False  False   False  False   True
8  False  False   False  False  False
9  False  False   False   True  False

Count of Missing Values in Each Column:
Name      0
Age       2
Gender    0
Marks     3
Grade     2
Ln: 57  Col: 0
```

```
IDLE Shell 3.12.10
File  Edit  Shell  Debug  Options  Window  Help
1  False  False   False   True  False
2  False   True   False  False  False
3  False  False   False  False   True
4  False  False   False   True  False
5  False   True   False  False  False
6  False  False   False  False  False
7  False  False   False  False   True
8  False  False   False  False  False
9  False  False   False   True  False

Count of Missing Values in Each Column:
Name      0
Age       2
Gender    0
Marks     3
Grade     2
dtype: int64

Dataset After Removing Rows with Missing Values:
     Name   Age  Gender  Marks Grade
0   Ayush  21.0    Male   85.0     A
6  Anjali  21.0  Female   95.0    A+
8   Sneha  22.0  Female   89.0     A

Dataset After Filling Missing Values:
     Name    Age  Gender      Marks          Grade
0   Ayush  21.00    Male  85.000000              A
1   Rahul  20.00    Male  85.142857             A+
2    Aman  20.75    Male  78.000000              B
3   Priya  21.00  Female  92.000000  Not Available
4    Neha  20.00  Female  85.142857              A
5   Rohan  20.75    Male  75.000000              B
6  Anjali  21.00  Female  95.000000             A+
7   Vivek  20.00    Male  82.000000  Not Available
8   Sneha  22.00  Female  89.000000              A
9   Karan  21.00    Male  85.142857              B
Ln: 57  Col: 0
```

Brief Explanation:

Identifies missing values using isnull(), counts missing values, removes incomplete rows, and fills missing numerical and categorical values using appropriate methods.

Q8.

Task 8 Merge, Concatenate, GroupBy & Pivot Table.py - C:/Users/ASUS/Desktop/internnova/week 2/Task 8 Merge, Concatenate, GroupBy & Pivot Table.py (3.12.10)

File Edit Format Run Options Window Help

```
import pandas as pd
df1 = pd.DataFrame({
    "ID": [1, 2, 3, 4],
    "Name": ["Ayush", "Rahul", "Aman", "Priya"],
    "Department": ["CSE", "ECE", "CSE", "ECE"],
    "Marks": [85, 90, 78, 92]
})
df2 = pd.DataFrame({
    "ID": [1, 2, 3, 4],
    "City": ["Delhi", "Jaipur", "Bhopal", "Mumbai"],
    "Age": [21, 20, 22, 21]
})
print("DataFrame 1:")
print(df1)
print("\nDataFrame 2:")
print(df2)
merged = pd.merge(df1, df2, on="ID")
print("\nMerged DataFrame:")
print(merged)
df3 = pd.DataFrame({
    "ID": [5, 6],
    "Name": ["Neha", "Rohan"],
    "Department": ["CSE", "ECE"],
    "Marks": [88, 75]
})
concatenated = pd.concat([df1, df3], ignore_index=True)
print("\nConcatenated DataFrame:")
print(concatenated)
grouped = df1.groupby("Department")["Marks"].agg(
    ["sum", "mean", "count", "min", "max"]
)
print("\nGroupBy Result:")
print(grouped)
pivot = pd.pivot_table(
    df1,
    values="Marks",
```

Ln: 7 Col: 2

Task 8 Merge, Concatenate, GroupBy & Pivot Table.py - C:/Users/ASUS/Desktop/internnova/week 2/Task 8 Merge, Concatenate, GroupBy & Pivot Table.py (3.12.10)

File Edit Format Run Options Window Help

```
})
df2 = pd.DataFrame({
    "ID": [1, 2, 3, 4],
    "City": ["Delhi", "Jaipur", "Bhopal", "Mumbai"],
    "Age": [21, 20, 22, 21]
})
print("DataFrame 1:")
print(df1)
print("\nDataFrame 2:")
print(df2)
merged = pd.merge(df1, df2, on="ID")
print("\nMerged DataFrame:")
print(merged)
df3 = pd.DataFrame({
    "ID": [5, 6],
    "Name": ["Neha", "Rohan"],
    "Department": ["CSE", "ECE"],
    "Marks": [88, 75]
})
concatenated = pd.concat([df1, df3], ignore_index=True)
print("\nConcatenated DataFrame:")
print(concatenated)
grouped = df1.groupby("Department")["Marks"].agg(
    ["sum", "mean", "count", "min", "max"]
)
print("\nGroupBy Result:")
print(grouped)
pivot = pd.pivot_table(
    df1,
    values="Marks",
    index="Department",
    aggfunc=["sum", "mean", "count", "min", "max"]
)
print("\nPivot Table:")
print(pivot)
```

Ln: 7 Col: 2

```
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/Task 8 Merge, Concatenate, GroupBy & Pivot Table.py
DataFrame 1:
   ID   Name Department  Marks
0   1  Ayush        CSE     85
1   2  Rahul        ECE     90
2   3   Aman        CSE     78
3   4  Priya        ECE     92

DataFrame 2:
   ID    City  Age
0   1   Delhi   21
1   2  Jaipur   20
2   3  Bhopal   22
3   4  Mumbai   21

Merged DataFrame:
   ID   Name Department  Marks    City  Age
0   1  Ayush        CSE     85   Delhi   21
1   2  Rahul        ECE     90  Jaipur   20
2   3   Aman        CSE     78  Bhopal   22
3   4  Priya        ECE     92  Mumbai   21

Concatenated DataFrame:
   ID   Name Department  Marks
0   1  Ayush        CSE     85
1   2  Rahul        ECE     90
2   3   Aman        CSE     78
3   4  Priya        ECE     92
4   5   Neha        CSE     88
5   6  Rohan        ECE     75

GroupBy Result:
            sum  mean  count  min  max
Department
CSE         163  81.5      2   78   85
ECE         182  91.0      2   90   92
```

```
DataFrame 2:
   ID    City  Age
0   1   Delhi   21
1   2  Jaipur   20
2   3  Bhopal   22
3   4  Mumbai   21

Merged DataFrame:
   ID   Name Department  Marks    City  Age
0   1  Ayush        CSE     85   Delhi   21
1   2  Rahul        ECE     90  Jaipur   20
2   3   Aman        CSE     78  Bhopal   22
3   4  Priya        ECE     92  Mumbai   21

Concatenated DataFrame:
   ID   Name Department  Marks
0   1  Ayush        CSE     85
1   2  Rahul        ECE     90
2   3   Aman        CSE     78
3   4  Priya        ECE     92
4   5   Neha        CSE     88
5   6  Rohan        ECE     75

GroupBy Result:
            sum  mean  count  min  max
Department
CSE         163  81.5      2   78   85
ECE         182  91.0      2   90   92

Pivot Table:
             sum  mean count   min   max
           Marks Marks Marks Marks Marks
Department
CSE          163  81.5     2    78    85
ECE          182  91.0     2    90    92
>>>
```

Brief Explanation:

Demonstrates combining DataFrames using merge and concatenation. GroupBy is used to calculate aggregate statistics, and a Pivot Table is created to summarize the data.

Q9.

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["Ayush", "Rahul", "Aman", "Priya", "Neha"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE"],
    "Marks": [85, 90, 78, 92, 88]
})

df["Result"] = df["Marks"].apply(lambda x: "Pass" if x >= 40 else "Fail")

print("Processed DataFrame:")
print(df)

df.to_csv("final_students.csv", index=False)

print("\nData exported successfully to final_students.csv")

exported_df = pd.read_csv("final_students.csv")

print("\nVerified Exported Data:")
print(exported_df)
```

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/ASUS/Desktop/internnova/week 2/Task 9 Exporting Data.py ==
Processed DataFrame:
    Name Department  Marks Result
0  Ayush        CSE     85   Pass
1  Rahul        ECE     90   Pass
2   Aman        CSE     78   Pass
3  Priya        ECE     92   Pass
4   Neha        CSE     88   Pass

Data exported successfully to final_students.csv

Verified Exported Data:
    Name Department  Marks Result
0  Ayush        CSE     85   Pass
1  Rahul        ECE     90   Pass
2   Aman        CSE     78   Pass
3  Priya        ECE     92   Pass
4   Neha        CSE     88   Pass
>>>
```

Brief Explanation:

Exports the processed DataFrame to a CSV file and reads the exported file again to verify that the processed data has been saved correctly.

Q10.

student_performance.csv

Name	Department	Marks	Result
Ayush	CSE	85	Pass
Rahul	ECE	90	Pass
Aman	CSE	78	Pass
Priya	ECE	92	Pass
Neha	CSE	88	Pass

Task 10 Mini Data Analysis Project.py - C:/Users/ASUS/Desktop/internnova/week 2/task 10/Task 10 Mini Data Analysis Project.py (3.12.10)

File Edit Format Run Options Window Help

```python
import pandas as pd
import numpy as np
df = pd.read_csv("student_performance.csv")
print("Dataset:")
print(df)
print("\nDataset Information:")
df.info()
print("\nFirst 5 Rows:")
print(df.head())
print("\nMissing Values:")
print(df.isnull().sum())
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
df["Department"] = df["Department"].fillna("Unknown")
print("\nDataset After Handling Missing Values:")
print(df)
print("\nSelected Columns:")
print(df[["Name", "Department", "Marks"]])
print("\nStudents with Marks Greater Than 80:")
print(df[df["Marks"] > 80])
print("\nStudents with Marks Greater Than 80 and Age Less Than 22:")
print(df[(df["Marks"] > 80) & (df["Age"] < 22)])
print("\nStudents Sorted by Marks:")
print(df.sort_values("Marks", ascending=False))
print("\nGroupBy Department:")
grouped = df.groupby("Department")["Marks"].agg(
    ["count", "mean", "min", "max", "sum"]
)
print(grouped)
print("\nPivot Table:")
pivot = pd.pivot_table(
    df,
    values="Marks",
    index="Department",
    aggfunc=["mean", "max", "min"]
)
```

Ln: 44 Col: 57

Task 10 Mini Data Analysis Project.py - C:/Users/ASUS/Desktop/internnova/week 2/task 10/Task 10 Mini Data Analysis Project.py (3.12.10)

File Edit Format Run Options Window Help

```python
print(df.isnull().sum())
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
df["Department"] = df["Department"].fillna("Unknown")
print("\nDataset After Handling Missing Values:")
print(df)
print("\nSelected Columns:")
print(df[["Name", "Department", "Marks"]])
print("\nStudents with Marks Greater Than 80:")
print(df[df["Marks"] > 80])
print("\nStudents with Marks Greater Than 80 and Age Less Than 22:")
print(df[(df["Marks"] > 80) & (df["Age"] < 22)])
print("\nStudents Sorted by Marks:")
print(df.sort_values("Marks", ascending=False))
print("\nGroupBy Department:")
grouped = df.groupby("Department")["Marks"].agg(
    ["count", "mean", "min", "max", "sum"]
)
print(grouped)
print("\nPivot Table:")
pivot = pd.pivot_table(
    df,
    values="Marks",
    index="Department",
    aggfunc=["mean", "max", "min"]
)
print(pivot)
print("\nNumPy Statistical Analysis:")
marks = np.array(df["Marks"])
print("Mean Marks:", np.mean(marks))
print("Maximum Marks:", np.max(marks))
print("Minimum Marks:", np.min(marks))
print("Standard Deviation:", np.std(marks))
df.to_csv("cleaned_student_performance.csv", index=False)
print("\nCleaned dataset exported successfully.")
```

Ln: 44 Col: 57

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
= RESTART: C:/Users/ASUS/Desktop/internnova/week 2/task 10/Task 10 Mini Data Analysis Project.py
Dataset:
     Name   Age  Gender Department  Marks
0   Ayush  21.0    Male        CSE   85.0
1   Rahul  20.0    Male        ECE   90.0
2    Aman   NaN    Male        CSE   78.0
3   Priya  21.0  Female        ECE   92.0
4    Neha  20.0  Female        CSE    NaN
5   Rohan  22.0    Male        ECE   75.0
6  Anjali  21.0  Female        CSE   95.0
7   Vivek  20.0    Male        ECE   82.0
8   Sneha  22.0  Female        CSE   89.0
9   Karan  21.0    Male        ECE    NaN

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        10 non-null     object
 1   Age         9 non-null      float64
 2   Gender      10 non-null     object
 3   Department  10 non-null     object
 4   Marks       8 non-null      float64
dtypes: float64(2), object(3)
memory usage: 532.0+ bytes

First 5 Rows:
    Name   Age  Gender Department  Marks
0  Ayush  21.0    Male        CSE   85.0
1  Rahul  20.0    Male        ECE   90.0
2   Aman   NaN    Male        CSE   78.0
3  Priya  21.0  Female        ECE   92.0
4   Neha  20.0  Female        CSE    NaN
```

Ln: 128 Col: 0

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Missing Values:
Name          0
Age           1
Gender        0
Department    0
Marks         2
dtype: int64

Dataset After Handling Missing Values:
     Name        Age  Gender Department  Marks
0   Ayush  21.000000    Male        CSE  85.00
1   Rahul  20.000000    Male        ECE  90.00
2    Aman  20.888889    Male        CSE  78.00
3   Priya  21.000000  Female        ECE  92.00
4    Neha  20.000000  Female        CSE  85.75
5   Rohan  22.000000    Male        ECE  75.00
6  Anjali  21.000000  Female        CSE  95.00
7   Vivek  20.000000    Male        ECE  82.00
8   Sneha  22.000000  Female        CSE  89.00
9   Karan  21.000000    Male        ECE  85.75

Selected Columns:
     Name Department  Marks
0   Ayush        CSE  85.00
1   Rahul        ECE  90.00
2    Aman        CSE  78.00
3   Priya        ECE  92.00
4    Neha        CSE  85.75
5   Rohan        ECE  75.00
6  Anjali        CSE  95.00
7   Vivek        ECE  82.00
8   Sneha        CSE  89.00
9   Karan        ECE  85.75

Students with Marks Greater Than 80:
     Name   Age  Gender Department  Marks
```

Ln: 128 Col: 0

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Students with Marks Greater Than 80:
     Name   Age  Gender Department  Marks
0   Ayush  21.0    Male        CSE  85.00
1   Rahul  20.0    Male        ECE  90.00
3   Priya  21.0  Female        ECE  92.00
4    Neha  20.0  Female        CSE  85.75
6  Anjali  21.0  Female        CSE  95.00
7   Vivek  20.0    Male        ECE  82.00
8   Sneha  22.0  Female        CSE  89.00
9   Karan  21.0    Male        ECE  85.75

Students with Marks Greater Than 80 and Age Less Than 22:
     Name   Age  Gender Department  Marks
0   Ayush  21.0    Male        CSE  85.00
1   Rahul  20.0    Male        ECE  90.00
3   Priya  21.0  Female        ECE  92.00
4    Neha  20.0  Female        CSE  85.75
6  Anjali  21.0  Female        CSE  95.00
7   Vivek  20.0    Male        ECE  82.00
9   Karan  21.0    Male        ECE  85.75

Students Sorted by Marks:
     Name        Age  Gender Department  Marks
6  Anjali  21.000000  Female        CSE  95.00
3   Priya  21.000000  Female        ECE  92.00
1   Rahul  20.000000    Male        ECE  90.00
8   Sneha  22.000000  Female        CSE  89.00
4    Neha  20.000000  Female        CSE  85.75
9   Karan  21.000000    Male        ECE  85.75
0   Ayush  21.000000    Male        CSE  85.00
7   Vivek  20.000000    Male        ECE  82.00
2    Aman  20.888889    Male        CSE  78.00
5   Rohan  22.000000    Male        ECE  75.00

GroupBy Department:
           count   mean   min   max     sum
```

Ln: 128 Col: 0

```
9  Karan  21.0   Male       ECE  85.75

Students Sorted by Marks:
    Name       Age  Gender Department  Marks
6  Anjali  21.000000  Female        CSE  95.00
3   Priya  21.000000  Female        ECE  92.00
1   Rahul  20.000000    Male        ECE  90.00
8   Sneha  22.000000  Female        CSE  89.00
4    Neha  20.000000  Female        CSE  85.75
9   Karan  21.000000    Male        ECE  85.75
0   Ayush  21.000000    Male        CSE  85.00
7   Vivek  20.000000    Male        ECE  82.00
2    Aman  20.888889    Male        CSE  78.00
5   Rohan  22.000000    Male        ECE  75.00

GroupBy Department:
            count   mean   min   max     sum
Department
CSE             5  86.55  78.0  95.0  432.75
ECE             5  84.95  75.0  92.0  424.75

Pivot Table:
             mean   max   min
            Marks Marks Marks
Department
CSE         86.55  95.0  78.0
ECE         84.95  92.0  75.0

NumPy Statistical Analysis:
Mean Marks: 85.75
Maximum Marks: 95.0
Minimum Marks: 75.0
Standard Deviation: 5.860887304837042

Cleaned dataset exported successfully.
>>>
```

Project Objective

The objective of this project is to analyze student performance data using NumPy and Pandas. The project includes data loading, inspection, missing-value handling, filtering, sorting, GroupBy analysis, Pivot Table creation, statistical analysis, and exporting the cleaned dataset.

Key Findings

- The dataset contains student performance information including **age, gender, department, and marks**.
- Missing values were identified and appropriately handled before analysis.
- Students scoring above 80 were identified through filtering.
- Department-wise performance was analyzed using **GroupBy**.
- Mean, minimum, and maximum marks helped compare student performance across departments.
- The Pivot Table provided a summarized view of marks for each department.
- NumPy was used to calculate statistical measures such as **mean, minimum, maximum, and standard deviation**.
- The processed dataset was successfully exported to a new CSV file.